

Risk of Dental Care with Hepatic Diseases

OMFS 512: Integrated Biomedical and Clinical Sciences

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Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Outcomes & Competency Statements

Learning Outcomes

1. Review the pathologies associated with liver failure.
2. Discuss diseases associated with the gallbladder.
3. Discuss the dental consideration and modification associated with hepatic disease.

Competency statements met

1.3, 1.5, and 2.1

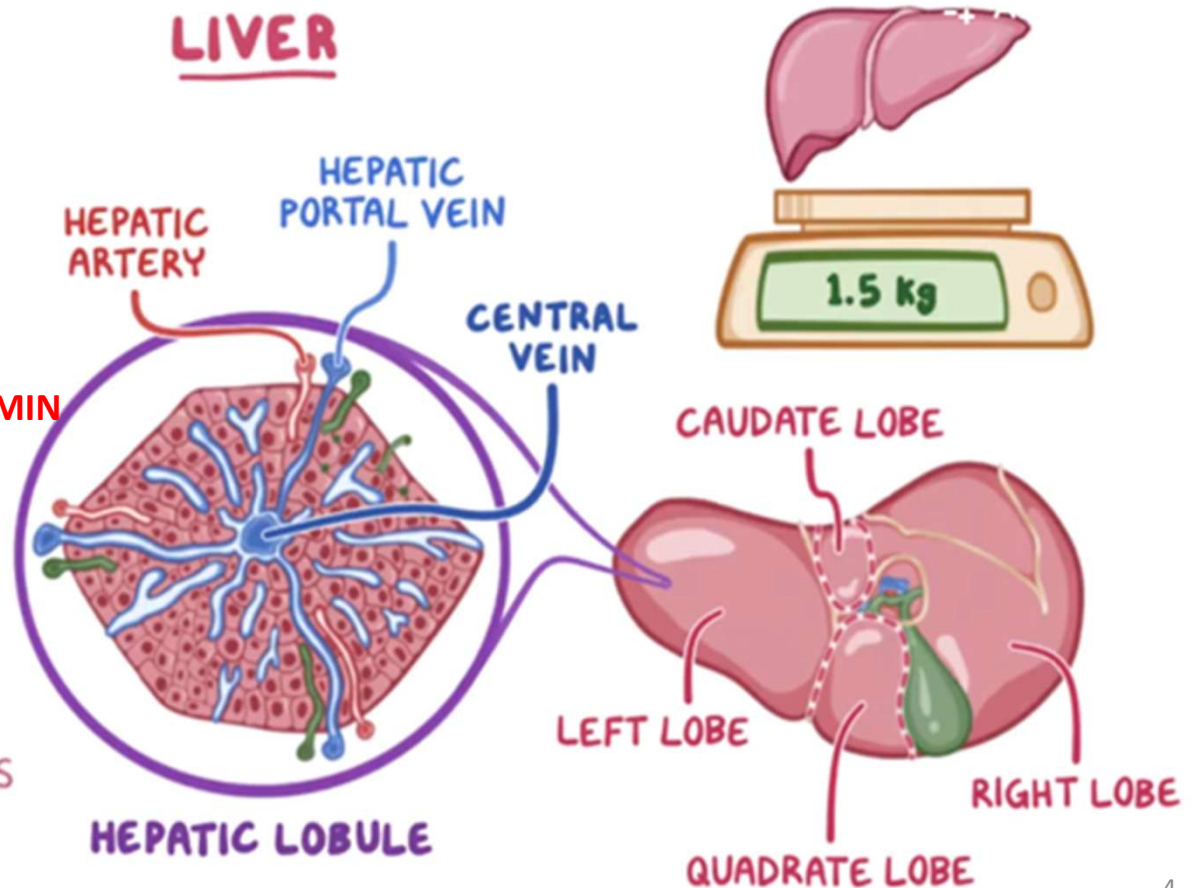
“Big Picture of Liver Disease”

- Liver is an organ that has over 200 functions
- Liver disease is common and can significantly affect the health of our patients.
- Dentists need to consider the following issues when treating patients with liver disease
 - **Excessive bleeding**
 - **Acquiring infections**
 - **Complications with drug metabolism: Local anesthetics and pain meds**
 - **Acquiring and transmission of Hepatitis B & C**
 - **End stage liver disease/hepatocellular carcinoma/liver transplantation**

Liver Anatomy and Function

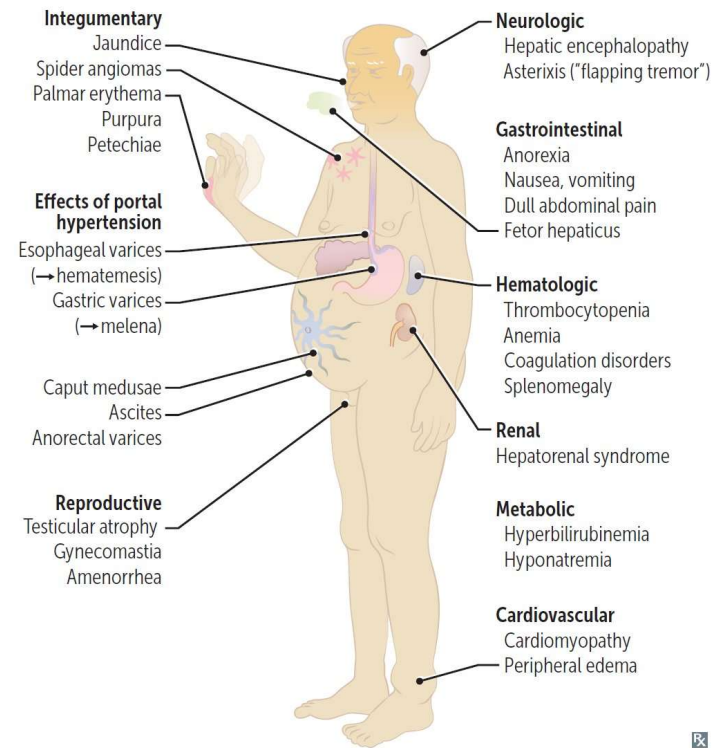
- * **LARGEST INTERNAL ORGAN**
- * **SURROUNDED** by a **CAPSULE** of **FIBROUS CONNECTIVE TISSUE** called **GLISSON'S CAPSULE**
- * **HEPATOCYTES PERFORM:**
 - ~ **PRODUCTION** of **BILE**, **PLASMA PROTEINS**, & **AMINO ACIDS**
 - ~ **METABOLISM** of **FAT**, **CARBOHYDRATE**, & **PROTEIN**
 - ~ **STORAGE** of **GLUCOSE**, **VITAMINS**, & **IRON**
 - ~ **BREAKDOWN** of **METABOLIC WASTE PRODUCTS**, **DRUGS**, & **TOXINS**

ALBUMIN



Clinical Manifestations of Liver Failure

- Jaundice
- Hypoglycemia
- Weight loss
- Peripheral edema
- Encephalopathy
- Prolonged/excessive bleeding
- Ascites
- Organomegaly



Cirrhosis - signs and symptoms

<http://www.betterfamilyhealth.org/liver-disease-diet.html>

Jaundice

- Jaundice is caused by the retention of bilirubin.
- Excessive bilirubin is deposited tissues
- Dark coloration of urine
- Pruritis (itchy skin)

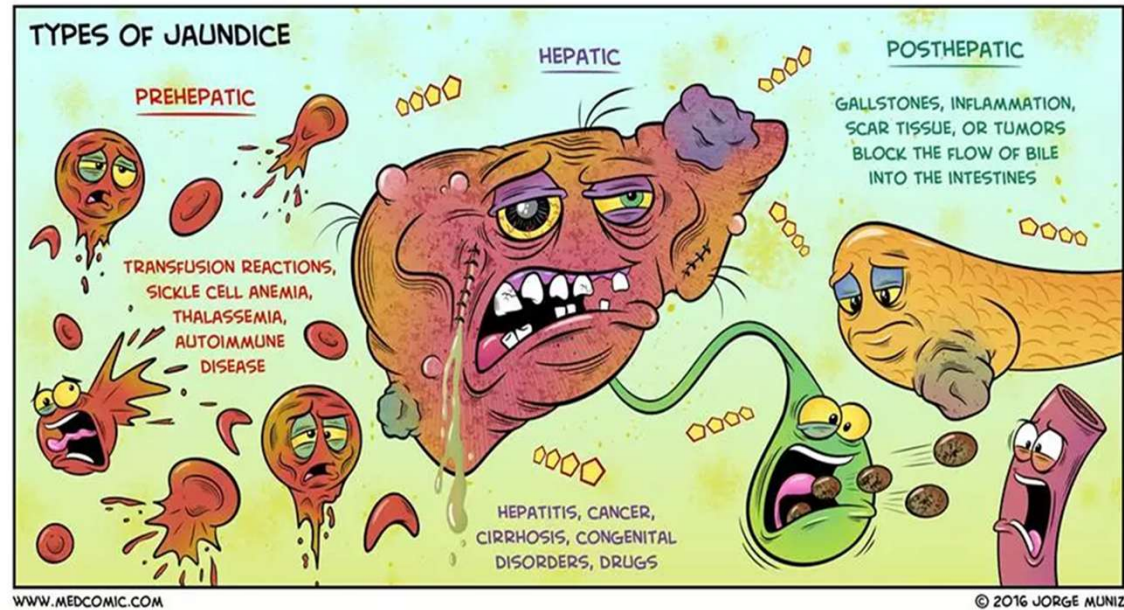


Tholey, D. *Jaundice* www.merckmanuals.com/professional/hepatic-and-biliary-disorders. Jan. 2023.

Jaundice

Types of Jaundice

- **Pre-hepatic**
 - Increased destruction of RBCs
- **Intrahepatic**
 - Liver disease
- **Post-hepatic**
 - Decrease in bile released from the liver due to blockage.



Peripheral Edema and Ascites



Amawasri Pakdara/Shutterstock

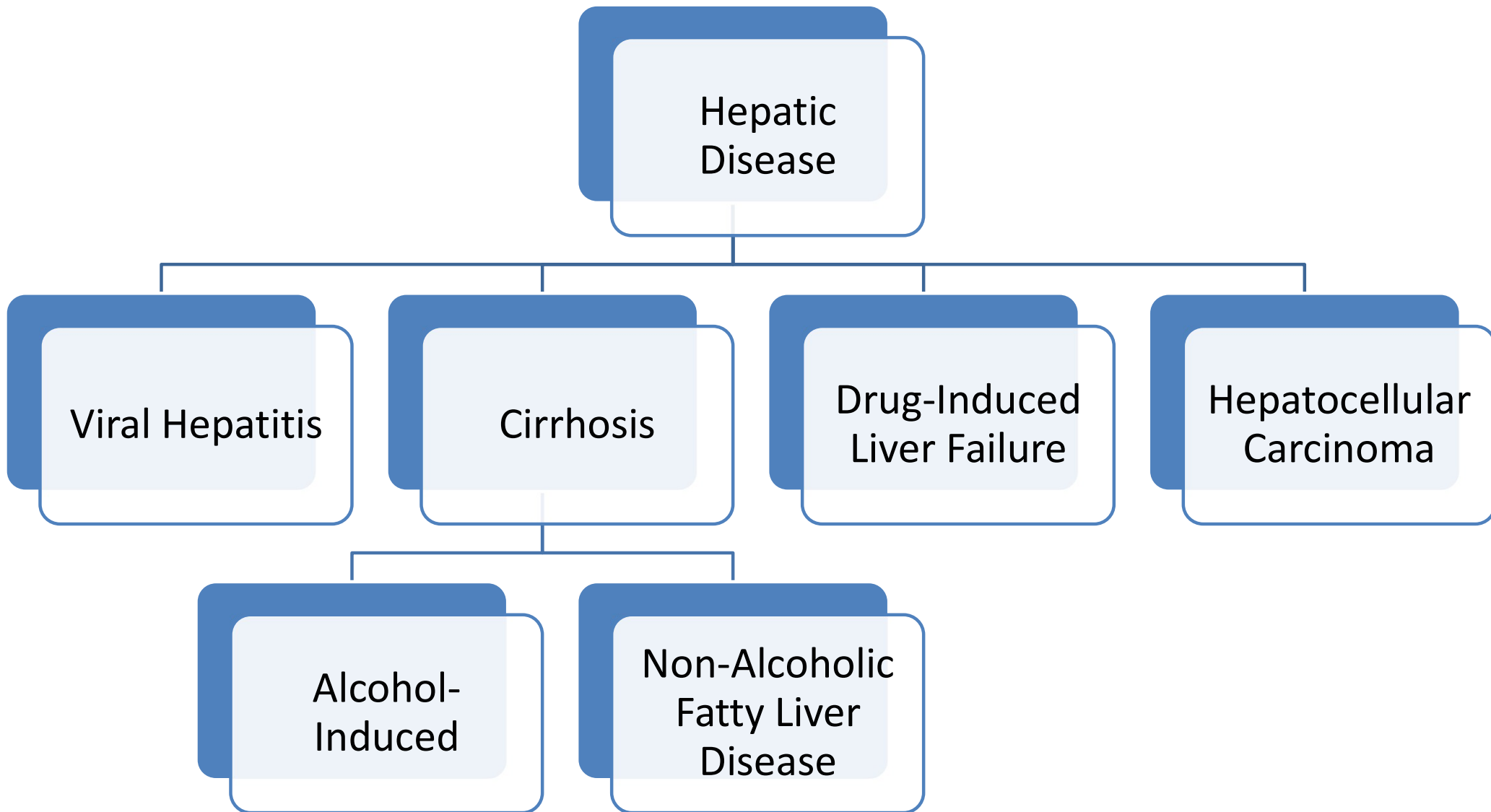


Fattyliverdisease.com

Vascular Findings



Wieland, C. (2016), Clinical examination: Skin. Clinical Liver Disease, 7: 119-125.



Viral Hepatitis

- Hepatitis A-E
- Acute or chronic liver disease
 - Hepatitis A&E are associated with acute liver disease
 - Hepatitis B, C, and D can be acute and become chronic
- Hepatitis B and C can cause hepatocellular carcinoma.
- Hepatitis B & C transmission

Table 6.1. Characteristics of Hepatitis Viruses: Pathogenesis/Etiology/Disease Course

	HAV	HBV	HCV	HDV	HEV
Classification	Picornavirus	Hepadnavirus	Flaviviridae family	Small defective RNA virus, infects with HBV	Calicivirus or alphavirus family
Mode of transmission	Fecal-oral, rarely: percutaneous	Percutaneous, sexual, perinatal	Percutaneous, rarely: sexual, perinatal	Percutaneous, sexual, perinatal	Fecal-oral
Prophylaxis	Ig, vaccine	HBIG, vaccine	None	None (HBV vaccine for susceptible)	None
Incubation days	15–50	30–180	15–160	21–140	14–63
<i>Clinical features</i>					
Chronic infection	No	1–10%, up to 90% in neonates	80–90%	Common	No
Carrier state	No	Yes	Yes	Yes	No
Severity of symptoms	Usually mild, age dependent	Moderate	Asymptomatic to mild	May be severe	Usually mild
Fulminant hepatitis	<0.1%	1%	Rare	Up to 20% in superinfection	10–20% in pregnant women
Hepatocellular carcinoma	No	Yes	Yes	?	No

Adapted from *Harrison's Manual of Medicine*; Fauci AS, Braunwald E, Kasper DL, et al. eds. 2009; New York, NY: The McGraw Hill Companies; 1264 pgs. Table 161-1.

HAV, hepatitis A virus; HBV, hepatitis B virus; HCV, hepatitis C virus; HDV, hepatitis D virus; HEV, hepatitis E virus; Ig, immune globulin.

Common symptoms of viral hepatitis



HEPATITIS A, B AND C



Diarrhea



Impaired appetite



Fever



Pain or discomfort in the abdomen

HEPATITIS D



Fatigue



Loss of appetite



Jaundice

HEPATITIS A, B AND C



Skin rash



Joint pain



Nausea



Vomiting



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Viral Hepatitis

- Prevention
 - Vaccines for hepatitis A&B
 - Following infection control policies
 - Standard Precautions
 - Avoid high-risk behavior
 - Good hygiene practices
 - Don't share needles
 - Practicing responsible sexual activity
- Treatment
 - Self-limiting
 - Supportive treatment to help with hydration
 - Medical treatment with antivirals
 - Transplant for chronic disease

Alcoholic liver disease

- Excessive consumption causes 60% of chronic liver disease in the Western world
- 40-50% of deaths from cirrhosis
- Over 10 million people suffer from alcoholism in the US
- Alcohol abuse 10th leading cause of death
- Increased risk of developing hepatocellular carcinoma

Cirrhosis

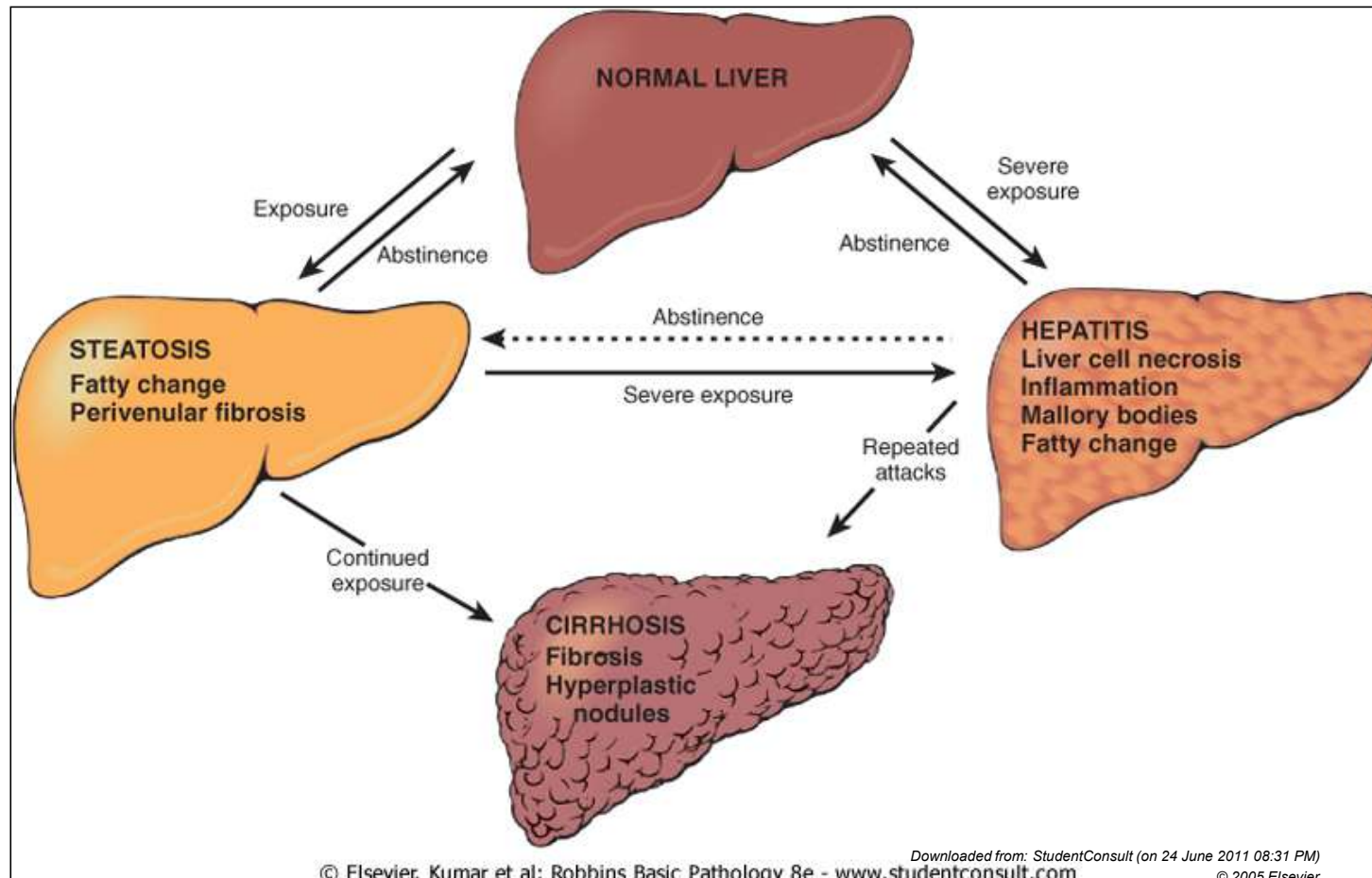
Characteristics of cirrhosis

1. Formation of fibrous bands and scars
2. Parenchymal nodules encircled by fibrous tissue
3. Disruption of the liver architecture



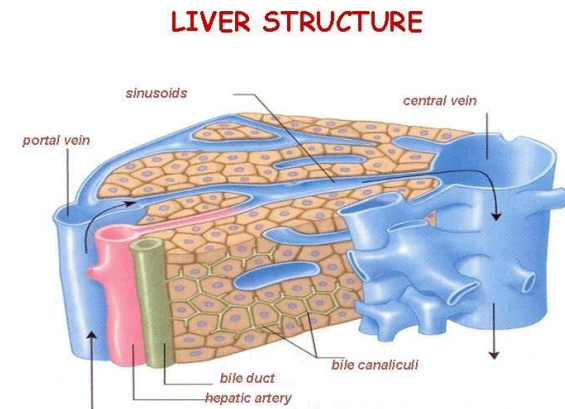
© Elsevier. Kumar et al: Robbins Basic Pathology 8e - www.studentconsult.com

Alcohol-induced Cirrhosis

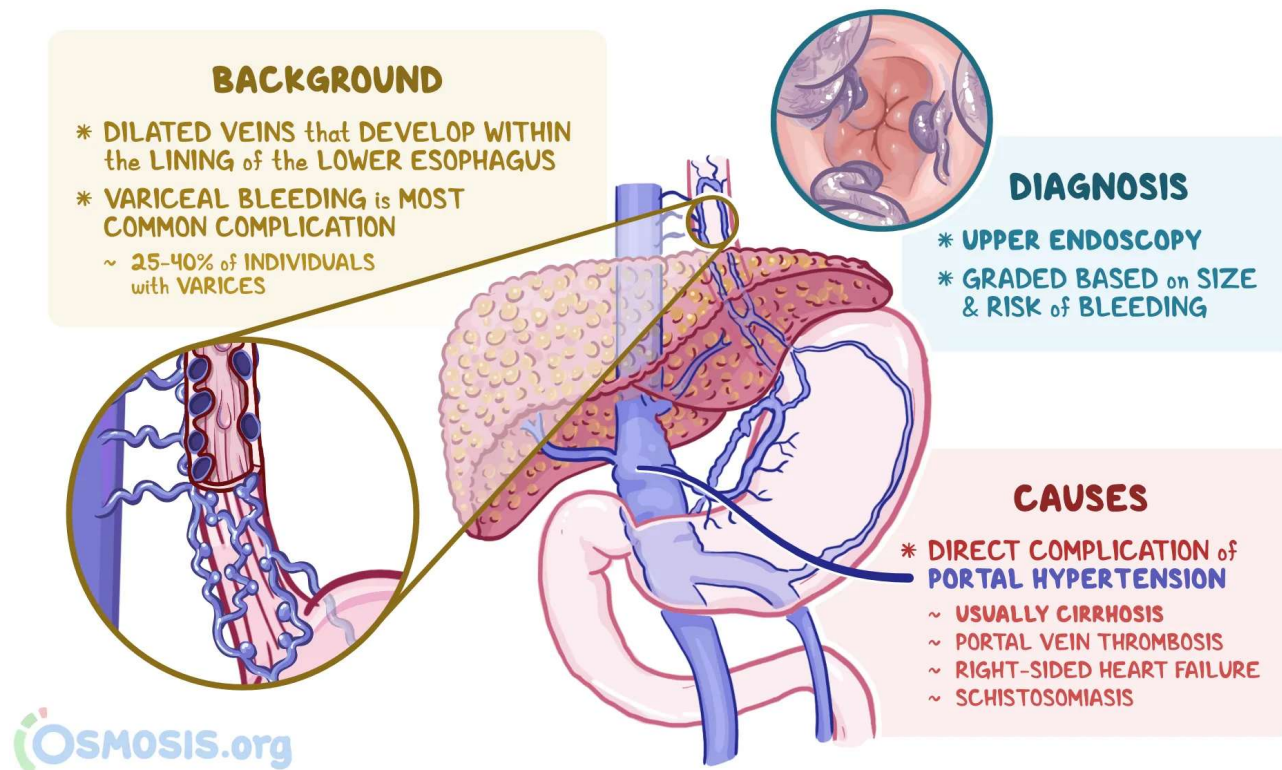


Portal hypertension

- Increased resistance to portal blood flow
- Dominant intrahepatic cause is cirrhosis.
- Fibrosis (deposited in perisinusoidal areas & around central veins) causes compression & resistance to portal blood flow.



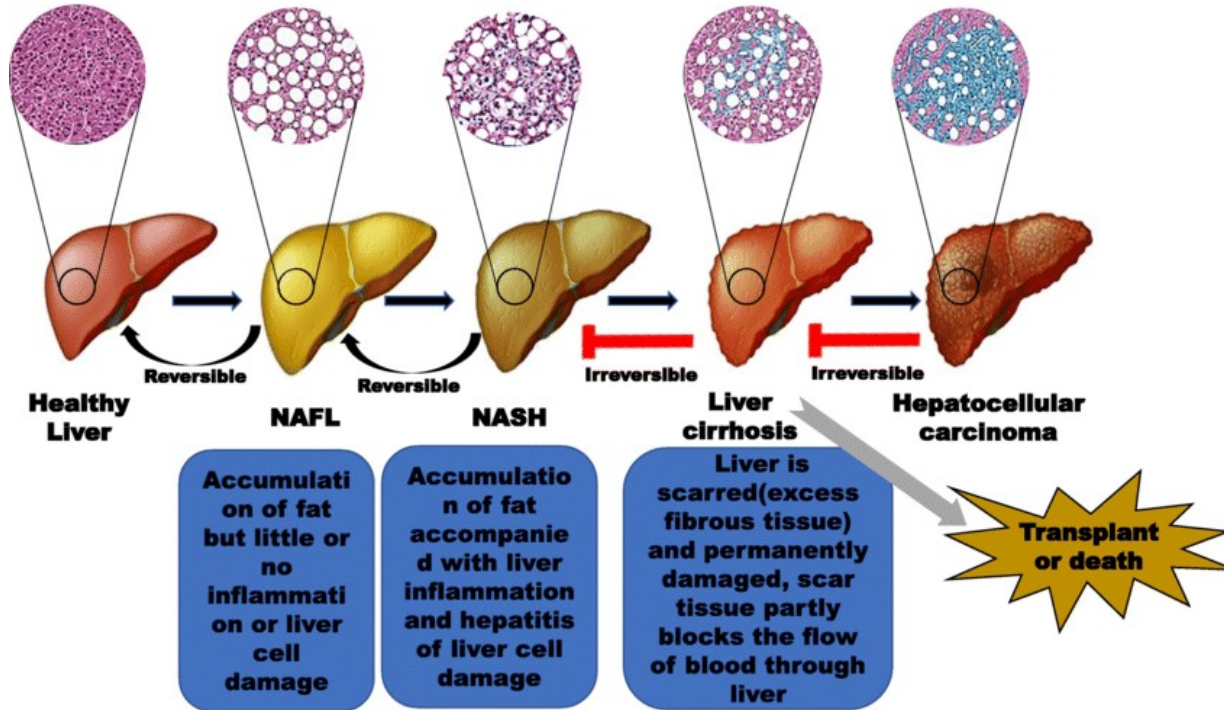
Esophageal Varices



Non-Alcoholic Fatty Liver Disease (NAFLD)

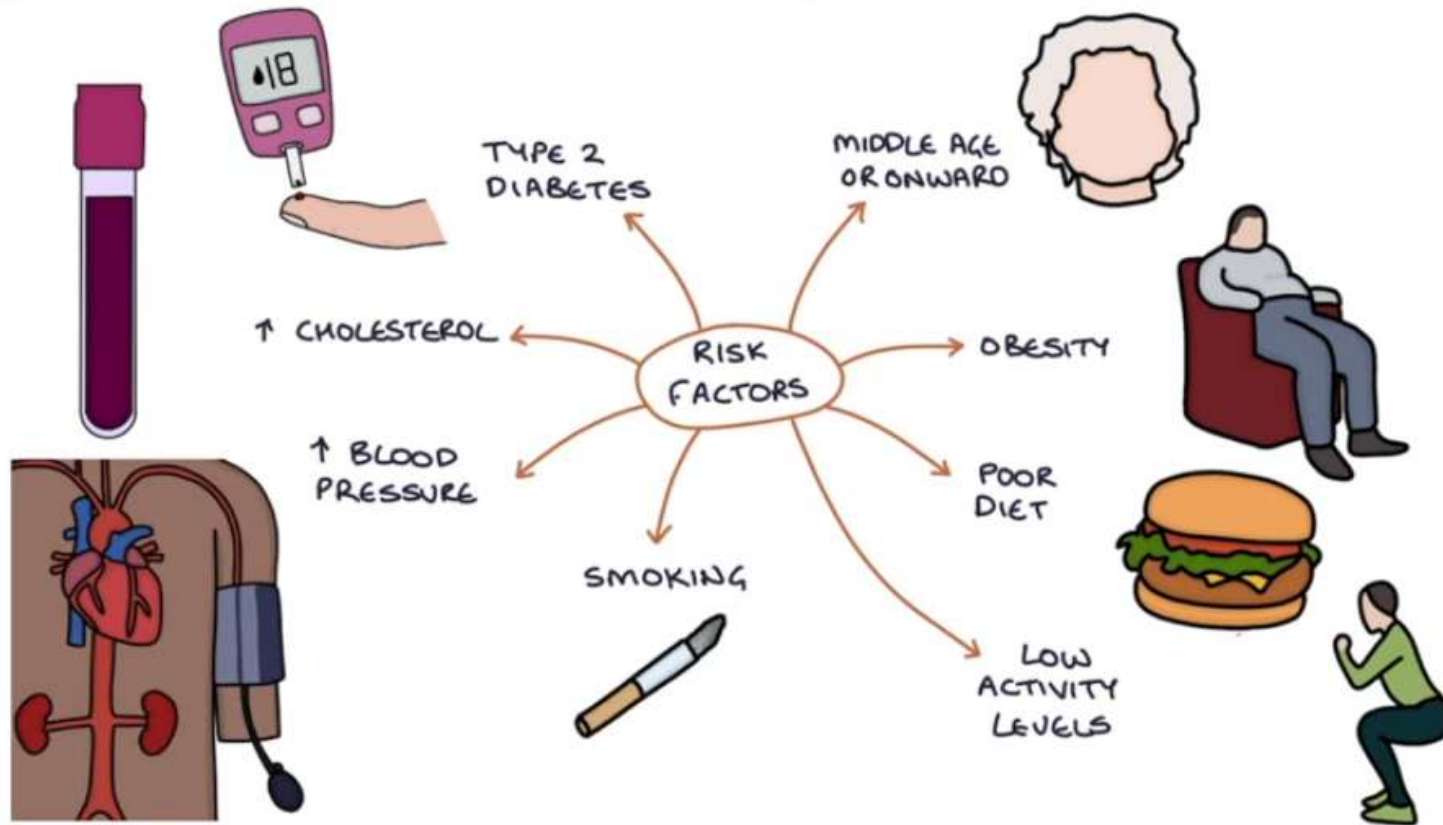
Excessive lipid storage in the liver NOT associated with alcohol consumption.

The spectrum of NAFLD

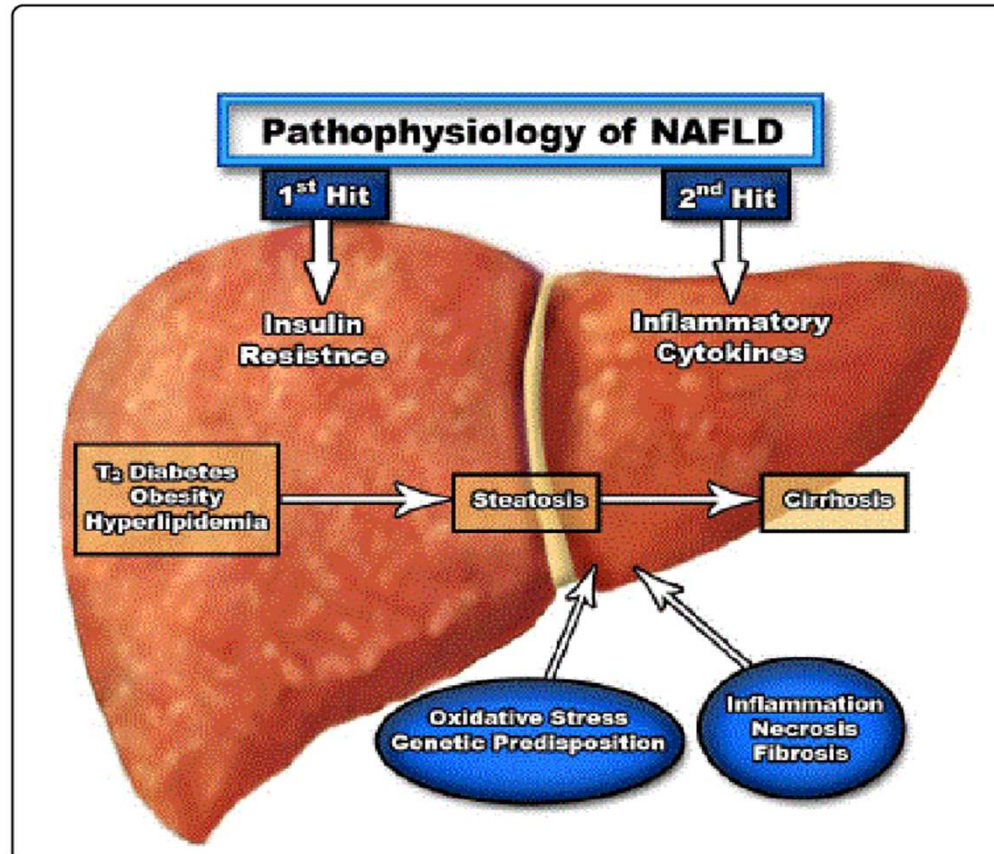


Xu L, Nagata N, Ota T. Int. J. Mol. Sci. 2019, 20, 5920

Risk Factors



Non-Alcoholic Fatty Liver Disease (NAFLD)



Clinical and Experimental Pharmacology 6(5):220

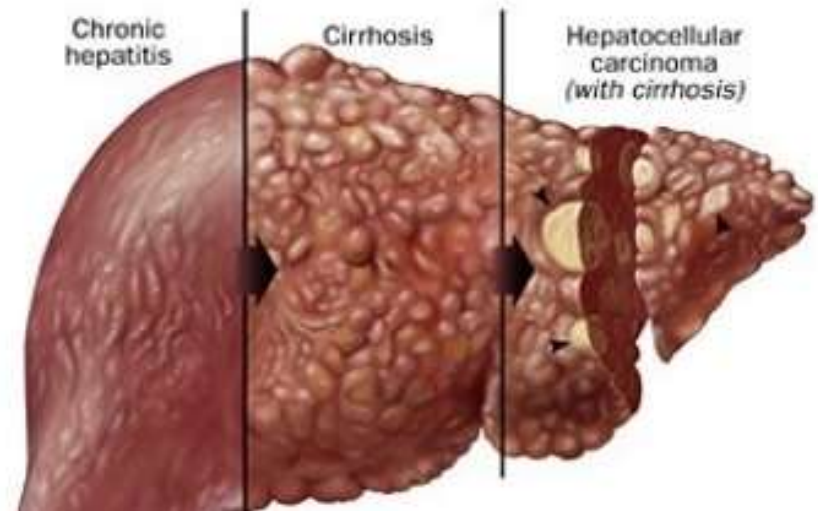
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Drug-induced liver disease

- Drugs and other toxins are detoxified in the liver.
- Acetaminophen is the most common cause of drug-induced acute liver failure
 - Classes of drugs which cause hepatotoxicity
 - Analgesics/Antipyretics - **Tylenol**
 - Anticancer drugs
 - Anesthetics
 - Antibiotics
 - Antituberculosis drugs
 - Antiretrovirals
 - Cardiac medications

Hepatocellular carcinoma

- Malignant disease of the liver
- Physical examination reveals a palpable mass
- Symptoms include:
 - Abdominal pain
 - Weakness
 - Weight loss
 - Increased risk of bleeding.



gastroinflorida.com/blog/hepatocellular-carcinoma-hcc

Liver Transplant

- Used to treat patient with end-stage liver disease
- Seen in those with
 - Autoimmune hepatitis (more common in females)
 - Drug-induced
 - Viral hepatitis
 - Malignancy
- Viral hepatitis & malignancy fare less well with transplant.

Cholestatic liver diseases

Cholestasis



Systemic retention of bilirubin, bile salts, and cholesterol.

Results from impaired bile flow

May present as jaundice

Bile salts can produce intense itching when deposited in skin.

Xanthelasma (cholesterol deposits, from impaired excretion)

Cholelithiasis “ Gall stones”

- Female predilection
- Occurs in adults, 25 years of age and up.
- Clinically silent 70% of the time
- Symptomatic: Stomach pain
- Gall bladder removal is done to resolve the disorder



www.thehealthsite.com/diseases-conditions/gear-up-against-gallstones-with-the-right-foods-kd0718-585162

Dental Considerations

- What liver disorder does the patient have? Acute vs chronic?
- Chronic liver disease, what stage?
- How is the disease being treated?
- What diseases may be associated with the liver disorder?
- Are there any medication the patient has been advised not to consume?
- During the examination, look for clinical signs of liver disease.
- Medical consultation for laboratory test
 - Liver function test: **Aspartate aminotransferase test (AST) and Alanine transaminase test (ALT), serum albumin, and serum alkaline phosphatase**
 - **Hepatitis Anti-body serum test**
 - **CBC w/diff**
 - **Bilirubin levels**

Dental Modifications

- 
- Infection control is key
 - Use of hemostatic agents to control bleeding
 - Conservative treatment, may require multiple appointments
 - Patients may have altered drug metabolism due to liver disease
 - Consultation with patient's physician(s)
 - Precautions with individuals with liver transplants
 - Patients are taking immunosuppressant medications
 - Be mindful of hepatitis B & C with disease transmission

Resources

